

The logo features the word "RAOH" in a large, bold, white sans-serif font. The background is a dark green with faint, large-scale geometric patterns, including a large "R" shape on the left and a series of thin, curved lines at the bottom right.

RAOH

— R e s e a r c h e r s —

RESEARCH. DISCOVER. IMPACT.

Advancing knowledge.
Creating real-world impact.

Cross-Sectional Study Writing Structure

How to Write a Scientific Paper (Simple Guide)

A cross-sectional study is usually written in four main sections:

- Introduction
- Methods
- Results
- Discussion

This structure is standard in medical journals and student research.

1. Introduction

Why are we studying this?

The Introduction explains the background of the problem and why the study is important.

What should it include?

- General background about the topic
 - Why the topic matters scientifically or clinically
 - What is already known (brief literature idea)
 - The research gap (what is missing)
 - The aim of the study
-

Simple idea

You are telling the reader: “Why did I do this study?”

Example

Topic: Sleep among medical students

You could say:

Medical students are known to experience high academic pressure, long study hours, and irregular sleep patterns. Previous studies have shown that poor sleep may affect cognitive performance and mental health. However, limited data exists regarding sleep quality among medical students in Jordan.

Therefore, this study aims to assess the prevalence of poor sleep quality among medical students and its associated factors.

2. Methods

How did we do the study?

This section explains exactly what you did so that another researcher can repeat your study.

What should it include?

1. Study design

Cross-sectional study

2. Population

Who you studied (e.g., medical students)

3. Sample

How many participants and how they were selected

4. Data collection tool

Questionnaire or validated scale (e.g., PSQI, PHQ-9)

5. Variables

What you measured (sleep, stress, GPA, etc.)

6. Statistical analysis

How you analyzed the data (SPSS, percentages, p-value)

Simple idea

You are telling the reader: “Exactly what I did and how I did it.”

Example

A cross-sectional study was conducted among medical students at a university. A total of 300 students were recruited using convenience sampling. Data were collected using a self-administered questionnaire, including demographic information and the Pittsburgh Sleep Quality Index (PSQI) to assess sleep quality.

Data were analyzed using SPSS software. Descriptive statistics were used to calculate frequencies and percentages. The association between sleep quality and academic performance was analyzed using Chi-square test, with a significance level set at $p < 0.05$.

3. Results

What did we find?

This section presents the findings without explanation or interpretation.

No opinions. No discussion. فقط أرقام ونتائج.

What should it include?

- Percentages
 - Means and standard deviations
 - Tables or graphs
 - Statistical significance (p-values)
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Simple idea

You are saying: “These are the results of my study.”

Example

Out of 300 participants, 180 students (60%) reported poor sleep quality.

There was a statistically significant association between sleep quality and GPA ($p = 0.02$), where students with poor sleep had lower academic performance.

4. Discussion

What do the results mean?

This is the most important intellectual part of the paper.

Here you interpret the results and connect them with other studies.

What should it include?

- Summary of main findings
 - Comparison with previous studies
 - Possible explanations
 - Implications (why it matters)
 - Limitations
 - Conclusion
-

Simple idea

You are telling the reader: “What do these results mean in real life?”

Example

This study found that a high proportion of medical students suffer from poor sleep quality. This finding is consistent with previous studies conducted in similar populations, which also reported high rates of sleep disturbances among university students.

The association between poor sleep and lower academic performance may be explained by reduced concentration, memory impairment, and increased fatigue.

However, this study has some limitations, including its cross-sectional design, which does not allow for establishing causality, and the use of self-reported data, which may introduce bias.

In conclusion, poor sleep quality is highly prevalent among medical students and is associated with reduced academic performance. Further longitudinal studies are recommended.

Quick Summary

Introduction

Why this study exists

Methods

How the study was done

Results

What was found (numbers only)

Discussion

What the results mean

Key Scientific Rule

Introduction explains the problem

Methods explain the process

Results show the data

Discussion explains the meaning

Discussion: The scientific method

1. What is the problem?

One question is: How can we prove something is true? For example, we can't prove that the Earth is round.